# Effect of steaming on physical-chemical properties and MtG performance of H-MFI zeolite catalyst

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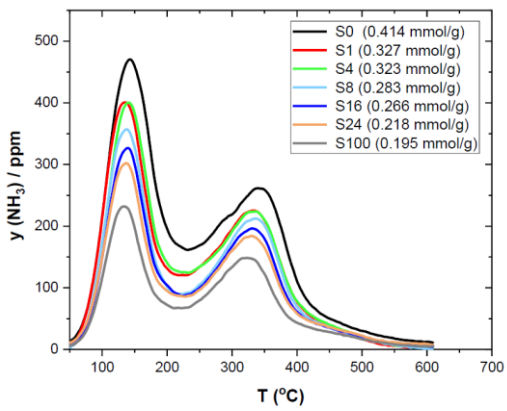
## Introduction

Renewable non-fossil origin fuels, or synthetic fuels, are the key to achieving carbon neutrality in the transport and mobility sectors. One of the most efficient methods for synthetic gasoline production is the methanol-to-gasoline (MtG) process. The renewable green methanol is used as a feedstock and converted into hydrocarbon fuels suitable for gasoline engines. The gasoline composition consists of various hydrocarbons, including paraffins, naphthenes, olefins, and aromatics. Adjusting process parameters and the properties of the zeolite-based catalyst through targeted acid strength adjustment can control the selectivity of these components. The resulting fuel composition impacts both the quality and yield of gasoline. In this work, the modification of zeolite catalysts by means of in situ steaming was conducted to increase the selectivity of the desired gasoline components.

## Experiment part

The steaming of zeolite catalysts was known as modification procedure to increase catalysts stability and its lifetime. This procedure causes dealumination, which leads to alterations in both the strength and distribution of the acid site. For the MtG catalysts, the effect of steaming on zeolites is mainly associated with changes in their acidity and structural properties.

The technical-shaped catalysts were steaming in situ at the temperature of 400 °C for different periods of time (1, 4, 8, 16, 24, and 100 hours). The samples' indexes refer to the steaming duration.




Firgure 1 – Specific amoung of acid and bacis sites

Figure 1 presents the $NH_3$-TPD for the steamed samples. The specific amount of acid and basic sites decreases with increasing steaming time, from 0.414 mmol/g for the reference catalysts to 0.195 mmol/g for the sample steamed for 100 hours.

The MtG tests were conducted at a temperature of 360 °C with time on stream 16 hours. With the higher steaming time and lower total number of acidic/bacsic sites the amount of aromatics components in organic liquid decreased for 10 wt.%. At the same time, the n- and iso-paraffins content increases from 31 to 37 wt.% without a drastic change of olefins. The detailed hydrocarbon analysis of the organic liquid product is presented in Figure 2.

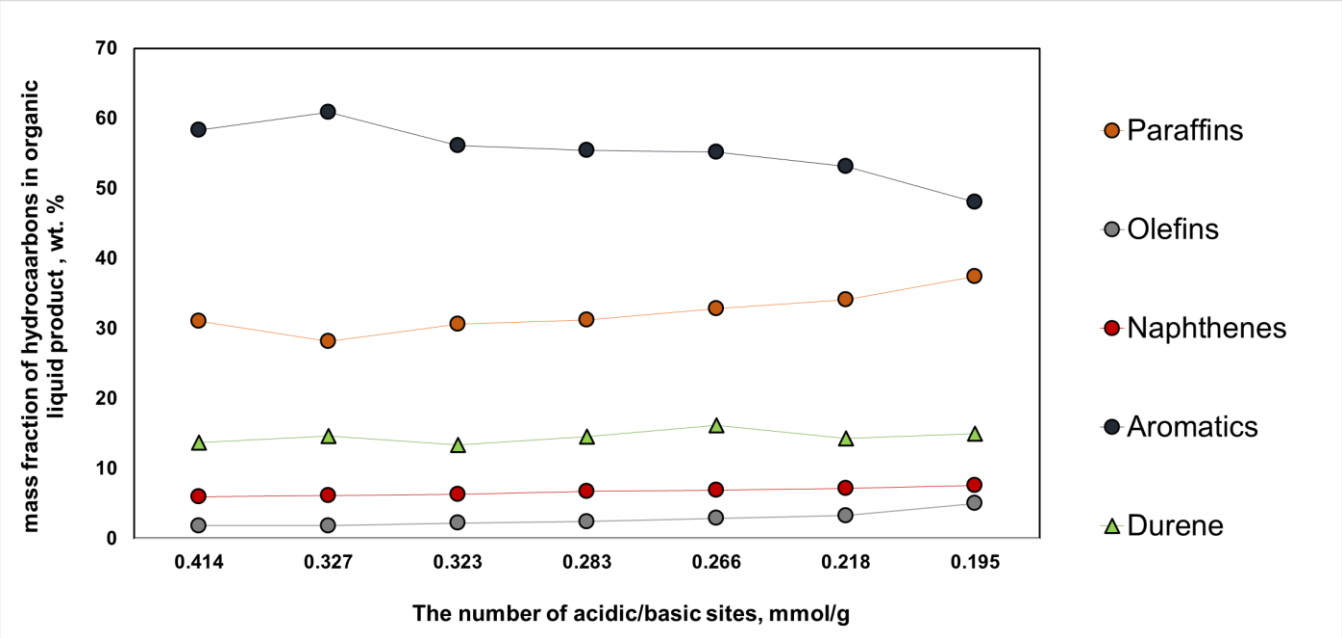



Figure 2 – detailed hydrocarbon analysis of the organic liquid product

**Conclusion and outlook**

The in situ steaming treatment may be a promising procedure for catalyst modification in advance of the MtG reaction. The time and amount of water are the main parameters of the steam treatment that affect the performance of the zeolite-based MtG catalysts:

- decrease of total acid and basic sites;
- in the product distribution lower aromatics and higher iso-paraffins and olefins.

Further analyses by means of BET and Al-NMR will be performed for the investigation of structural and morphological changes during the steaming. Additionally, the later experiments will focus on steaming temperature influence, distinguishing the catalyst acid and basic sites distributions as well as the determination of the zeolite structural changes in order to understand the mechanism of MtG catalyst steaming.